



Lock 57 Park
Ohioville, Pa.
Humming Bird & Butterfly Garden

Dr. Vincent Troia -
Chief Executive Chairman & CEO,
Co-founder
Ohio River Trails Council

John Szatkiewicz -
Mayor, Ohioville Boro
Midland, Pa.

Karen Atkins -
Owner, Proper Gardens
New Brighton, Pa.

Pam Varga -
Master Gardener
Penn State Extension
Beaver County

Dino De Ciantis -
Ed. D.
Director, Penn State Center
Penn State University

The Garden Site

The humming bird and butterfly garden at Lock 57 Park will provide habitat for humming birds and butterflies, as well as other pollinators and wildlife. From early spring through late fall, park visitors will be able to enjoy the beauty of the plants and flowers in this special garden. The original site suggested was a 10' x 100' plot located along a steep hillside near the park entrance. This site presented a number of problems:

- * A garden on a steep, rocky hillside would be difficult to plant and maintain.
- * Terracing the slope would remove valuable topsoil.
- * Initially, there would be erosion on the bare slopes above and below the terrace.
- * Surrounding vegetation which should be left in place to prevent erosion would eventually creep into the garden and overtake it.
- * There are currently wild flowers including Blood Root, Spring Beauties, and Dead Nettle growing on the site.
- * Visitors could view the garden, but would not be able to walk into it.

A second site, a grassy slope on the right side of the entrance road was considered next. Visitors would drive by a garden here, rather than interact with it. Since the area is sloped, terracing or raised beds would again be necessary.

A final site near the center of the park was chosen. This grassy flat plot would be easier to prepare, plant and maintain and would be more accessible for visitors who could enjoy walking through the garden. In this location, the garden would provide a focal point and compliment the existing park structures.

An important consideration in maintaining the garden is the lack of water within the park. Water for plants must be carried by hand from the river which is nearby.

Ohioville Butterfly Preserve

Design Overview:

The original concept included a 10 x 100 ft. ribbon of perennials, shrubs, biennials and annuals to attract butterflies.

The specified location was on a hillside close to the parking area, due to concern that foot traffic could endanger the garden.

A Master Gardener (Pam Varga) and professional garden designer (Karen Atkins) agreed that erosion, encroaching weeds, and the removal of top soil necessary to establish a planting "shelf" on the hillside made it a poor choice for the garden area. In addition, long-term maintenance would be daunting on a slope.

To achieve the original goal of attracting butterflies without leaving plants vulnerable to traffic, we suggest grouping the plants in one enclosed space, for more impact. The area can accommodate seating for guests, and should be enclosed by fencing to discourage abuse. Arbors and fences will provide vertical support for honeysuckle, clematis, and other plants that are fragrant, colorful, and favorites of butterflies. Encroaching weeds would not be a concern.

For this reason, we are submitting a slightly different design that was originally requested. We hope the committee finds it of use, but would be happy to change course if requested.



Garden Sites

- A. Original proposed site - 10" x 100" hillside left of the park entrance
- B. Second proposed site - Grassy slope to the right of the park entrance
- C. Final site - flat, sunny area between picnic shelter and restrooms near center of the park
- D. Water source - river - final site is to the right in this photo

A



B



C



D



Plant Selection and Design

A design was chosen from three alternatives. Plants will be contained in raised beds. An arbor will mark the main entrance. Paths through the garden will be covered with pea gravel and will be handicapped accessible.

Plants were selected according to the following criteria:

1. Plants which provide nectar for humming birds and butterflies
2. Host plants which provide food for butterfly larva
3. Trees and shrubs to provide nectar, larval food, shelter, nesting space, and shade
4. Bloom time for plants, beginning in early spring and continuing through summer until the first frost
5. Use of Native perennials
6. Suitability to the site
7. Plant diversity
8. Aesthetic qualities - color, height, texture, habit
9. Project budget and plant prices
10. Availability from the supplier

Because the suggested design is subject to project funds and budgeting, plant prices, and availability of plants from the supplier, substitutions may be made or quantities may be changed.

The Garden

At the main entrance to the garden, visitors will pass through an arbor covered with vibrant orange Honeysuckle vines, a favorite of humming birds. In the front beds, Butterfly Bush and bright red Penstemon and Cardinal Flower are humming bird and butterfly magnets. Shrubs near the other entrances will be a source of nectar early in the spring and larval food later in the summer. Shrubs also provide shelter and nesting sites. A variety of brightly colored perennials in the other beds provide nectar for humming birds and butterflies as well as food for hungry caterpillars throughout the summer until frost. Columbine will be a source of nectar for butterflies and humming birds early in the spring. Asters are a particularly good source of nectar from late summer into the Fall. A dogwood tree in the center of the garden provides nectar from spring blossoms, food for butterfly larva, shelter, and nesting sites. Fragrant herbs planted in the center bed are host plants for a variety of butterfly larva. Walking the paths through the garden, park visitors can easily enjoy the beauty of the garden and the park.

Final Design and Construction

Two alternate garden designs are based on the final site selection and plant purchase. The site, a flat, sunny area between the picnic shelter and the restroom, will provide a focal point near the center of the park. Visitors will have easy access to the garden and be able to enjoy walking through it. Plants will be contained in raised beds. Raised beds will be easier to maintain and prevent encroachment by surrounding vegetation. The designs differ in the way the plant beds are shaped in the center of the garden. In one design, the beds are notched, and in the other design, the beds are diagonal. Suggested plant placement in both designs is the same.

After measuring and staking the site, the soil should be lightly tilled to avoid underground drainage pipes. The soil should then be raked to remove grass and root clumps. The raised beds could be constructed of 2 x 10 pressure treated lumber or any other weather resistant wood. The timbers should be placed 2 - 3 inches in the ground resulting in beds about 7 - 8 inches high. The beds would then be filled with garden soil. Plants should be placed according to the planting distances included in the chart and receive plenty of water. During the first few weeks, plants should be watered frequently to help them establish strong root systems. Watering may be difficult as the only source is the nearby river.

An arbor constructed with weather resistant lumber will be placed at the garden entrance. Honeysuckle vines will be planted on either side so that in time, the vines can grow up over the arbor.

Pathways should be covered in pea gravel or other similar porous material to allow for drainage.

Resources - Web Sites Humming Bird & Butterfly Gardens

1. <http://allegheny.extension.psu.edu/Horticulture/PSMG/PollinatorProject.html> (Also see the links given)
2. <http://ento.psu.edu/pollinators> (Click on "Certify your pollinator friendly garden with Penn State Master Gardeners")
3. www.thegardenhelper.com/hummingflower.html
4. www.njaudubon.org/sectionBackyardHabitat/CreateaGarden.aspx
5. www.OhioRiverTrail.org (Click on "Butterfly Preserve")

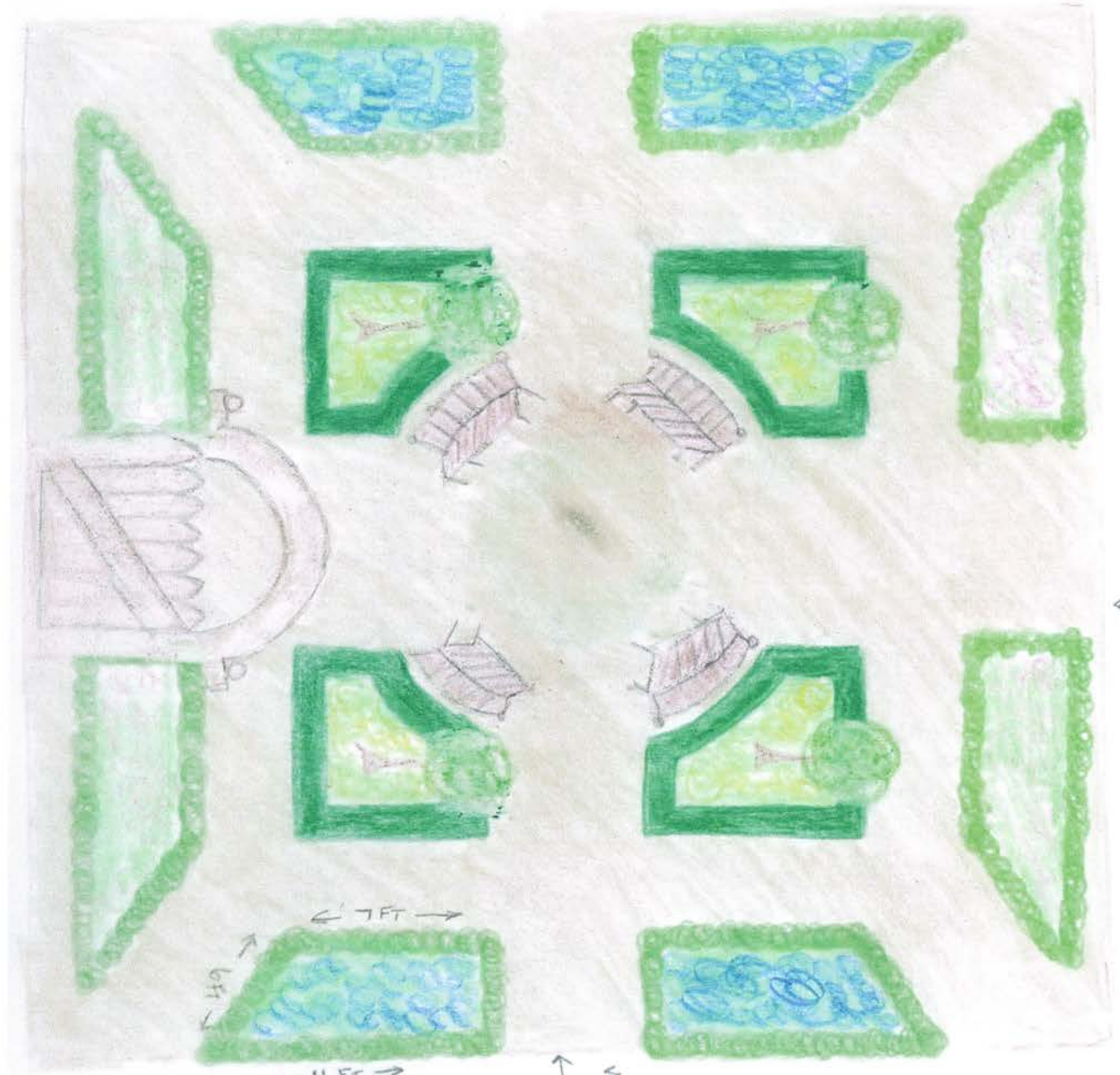
Lock 56 Humming Bird and Butterfly Garden Perennial Flowers

Common Name	Scientific Name	Quantity (Alternate)	Bloom Time	Color	Planting Distance
Butterfly Weed	Asclepias tuberosa	6	Summer - Fall	Orange -	2 ‘
Blue Lobelia	Lobelia siphilitica	10 (8)	Summer- Fall	Blue	12”
Goldenrod	Solidago	3	Summer - Fall	Yellow	2’
Blazing Star	Liatris spicata	5	Summer	Purple	12”
Aster	Aster novae-angliae	3	Summer - Frost	Purple	18” – 2’
Perennial Sunflower	Helianthus	8 (6)	Summer - Fall	yellow	2’
Beard Tongue	Penstemon barbatus	8 (6)	Summer	Red	14”
Cardinal Flower	Lobelia cardinalis	8 (6)	Summer - Fall	Red	12”
Anise Hyssop	Agastache foeniculum	3	Summer - Fall	Purple	20”
Columbine	Aquilegia canadensis	8	Spring	Various	18”
Monarda	Monarda didyma or Monarda fistulosa	6	Summer	Red, purple, or pink	14” – 18”
Black Eyed Susan	Rudbeckia hirta	6	Summer - Fall	Yellow	12”
Coneflower	Echinacea	5 (3)	Summer - Frost	Purple, various	16”
Herbs					
Thyme		8 (6)			Host plant
Oregano		8 (6)			Host plant
Parsley					Host plant
Dill					Host plant



Lock 56 Humming Bird and Butterfly Garden Plant List - Vines, Trees, Shrubs

[illegible]



PEA GRAVEL

WHEELCHAIR & VISIBILITY ACCESSIBLE

40 x 90 Ft Enclosure



Access for
Wheelchairs
and
Wheelbarrows

Finals
could be
crafted by
local artisans

40 x
40 ft
ENCLOSURE

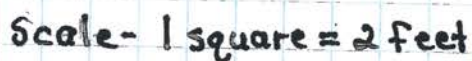
ARBO

3x3 ft PAVERS
ALTERNATING
w/ CREEPING VILLO



W
A
W
A

Humming Bird Lock 57 Park & Butterfly Garden



$\leftarrow 40 \rightarrow$



Trees-

DW - Dogwood

RB- Redbud



Shrubs -

AZ - Azalea

BB-Butterfly Bush

L- Lilac

SB- Spruce bush



H-Honeysuckle

Perennials

Ah - Anise Hyssop

C1 - Cone flower

M-Monarda

As - Aster

Co - Columbine

ps - Penstemon

At - Butterfly Weed

G - Golden rod

R- Rudbeckia

B1 - Blue Lobelia

He- Helianthus

--	--	--	--	--	--

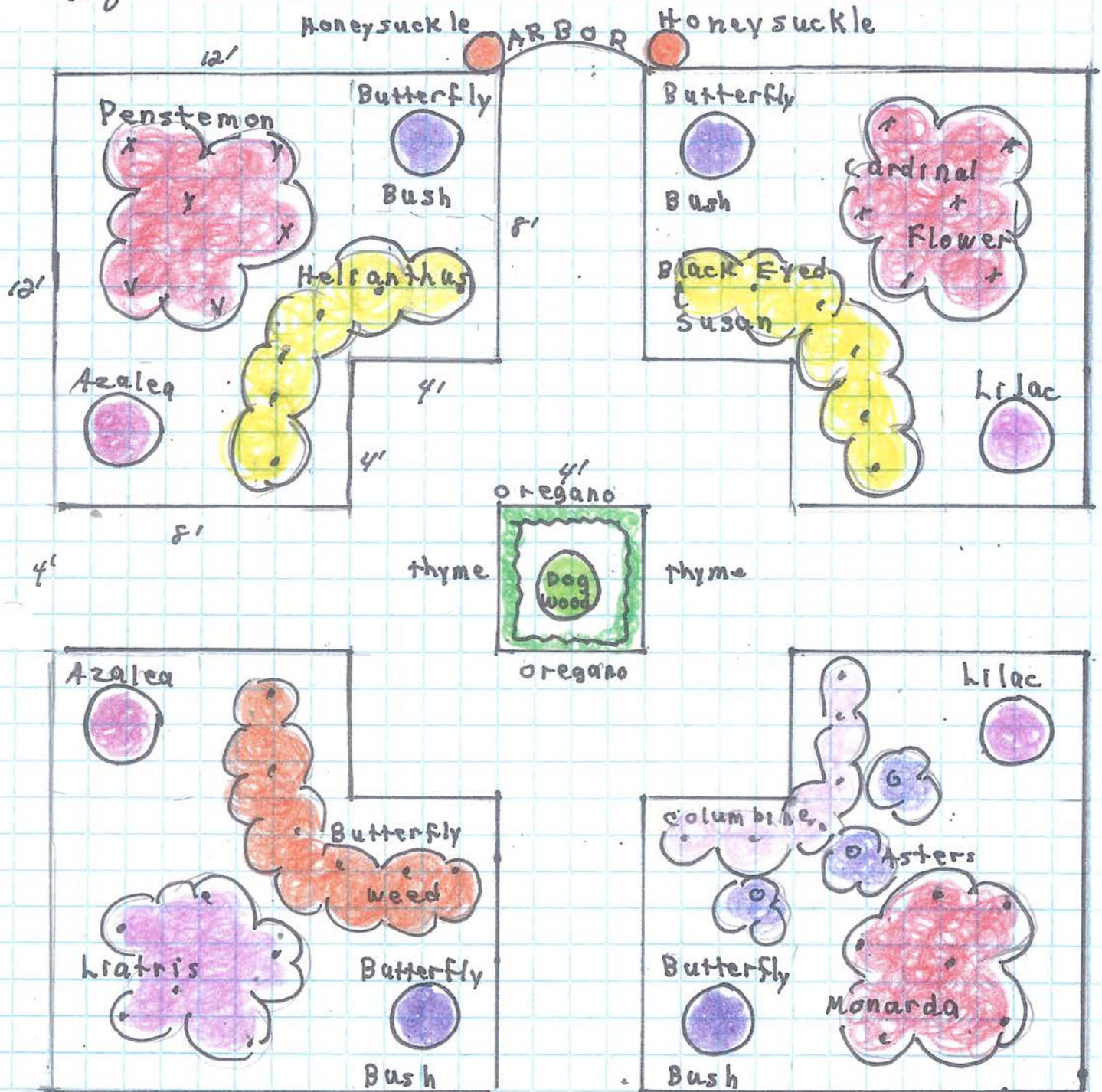
CF - Cardinal Flower

$$h_j - h_{j+1} \text{ atris}$$

5-16-11

Lock 57 Hummingbird & Butterfly Garden Park

1 square = 1 foot

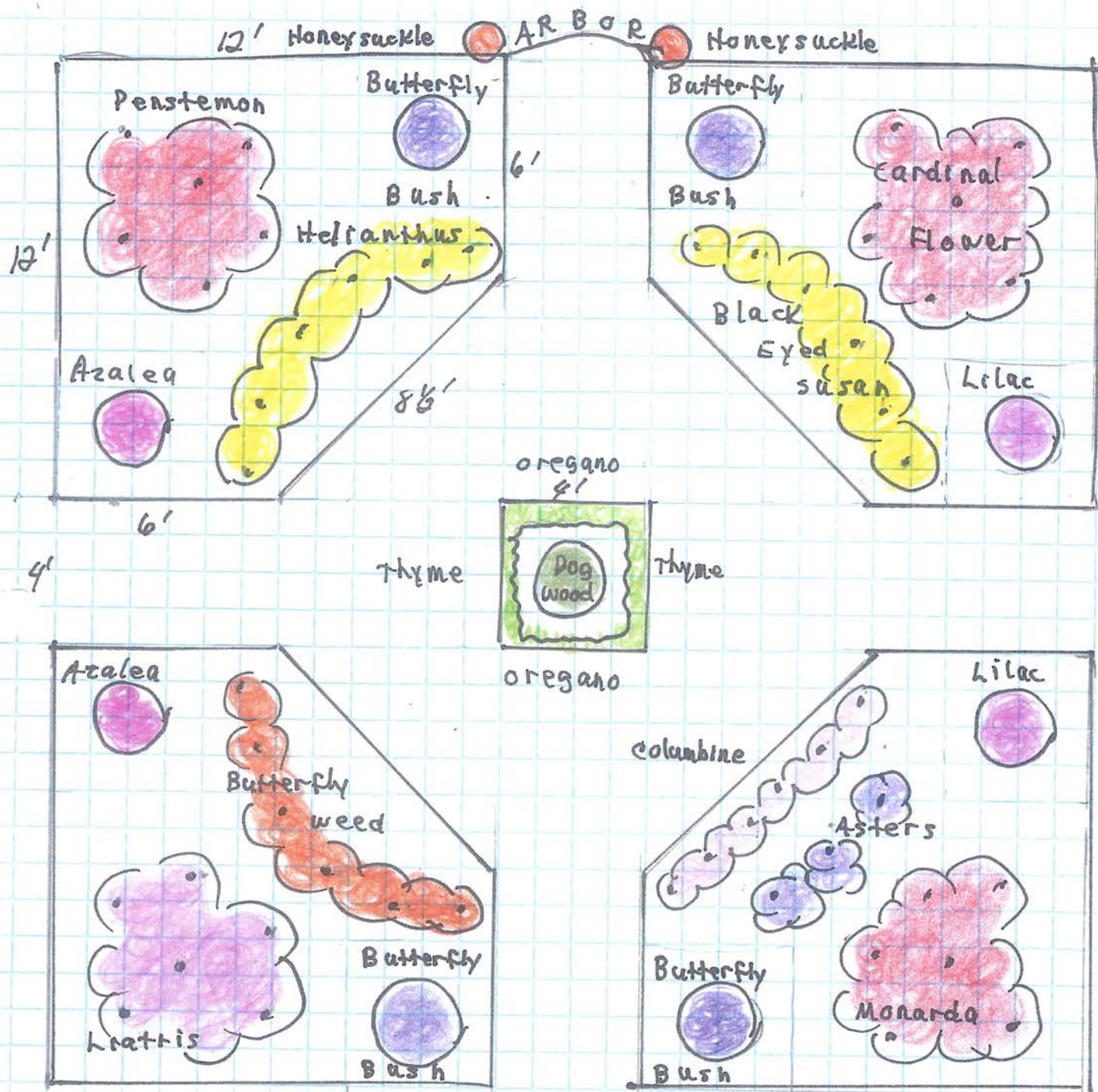


- [Some colors may vary]
- | | | | | |
|-----------------------|------------------|--------------------------|---------|----------|
| 1 Dogwood | 4 Butterfly Bush | 2 Honeysuckle | 2 Lilac | 2 Azalea |
| 6 Butterfly weed (2') | | 6 Cardinal Flower (12") | | |
| 6 Liatris (12") | | 6 Columbine (18") | | |
| 3 Asters (18") | | 6 Monarda (18") | | |
| 6 Helianthus (2') | | 6 Black Eyed Susan (12") | | |
| 6 Penstemon (12") | | 6 Coneflower (18") | | |
| (Planting Distance) | 8 Thyme | 8 Oregano | | |

Lock 57 Hummingbird & Butterfly Garden

5-16-11

1 square = 1 foot



Colors may Vary

- | | | | |
|-----------------------|--------------------------|---------------|---------|
| 1 Dogwood | 4 Butterfly Bush | 2 Honeysuckle | 2 Lilac |
| 6 Butterfly Weed (2') | 6 Cardinal Flower (12') | | |
| 6 Liatris (12') | 4 Columbine (18') | | |
| 3 Asters (18') | 6 Monarda (18') | | |
| 6 Helianthus (2') | 6 Black Eyed Susan (12') | | |
| 6 Penstemon (12') | 6 Coneflower (12') | | |
| C Planting Distance) | 8 Thyme | 8 Oregano | |

Land Of
United States Of America
D.B. Vol. 865 Pg. 205
TP# 71-170-112.4
Tract No. 1925
Area = 11.703 Acres

Land Of
S.H. Bell Company
D.B. Vol. 1099 Pg. 784
Parcel No. 1
TP# 71-170-112

Land Of
United States Of America
D.B. Vol. 878 Pg. 224
TP# 71-170-112.3
Tract No. 1924
Area = 1.163 Acres

PROPOSED ELECTRIC SERVICE
IN EXISTING PAVILION

PROPOSED DUSK TO DAWN LIGHT
IN CONCRETE TOILET BUILDING

PROPOSED POLE

PEMIG Wetlands
Reference: National Wetlands Inventory Map,
U. S. Department of the Interior Dated 1988,
East Liverpool North, Ohio - PA. - W. Va. Quad

PORT OF PITTSBURGH GRANT PROPOSAL

BUTTERFLY/HUMMINGBIRD PROPOSAL

KAYAK/BIKE RACK PROPOSAL



CALL BEFORE YOU DIG!
PENNSYLVANIA LAW REQUIRES
3 WORKING DAYS NOTICE FOR
CONSTRUCTION PHASE AND 10 WORKING
DAYS IN DESIGN STAGE - STOP CALL
Pennsylvania One Call System, Inc.
1-800-242-1776

DANIEL C. BAKER ASSOCIATES, INC.
CONSULTING ENGINEERS, SURVEYORS AND PLANNERS
6056 TUSCARAWAS ROAD
BEAVER, PENNSYLVANIA 15009
Phone: (724) 495-7020 Fax: (724) 495-2594 www.dcbaker.com

REVISIONS

DATE	DESCRIPTION

Date: January 28, 2011
Scale: 1" = 40'
Drawn: TJS
Checked: TJS
Reviewed: ---
Field Book/Per: ---
Drawing No.: ---
Job No.: 00100-1102
CAD File: 00100110201
Sheet No.: 1 of 1

OHIOVILLE BOROUGH
LOCK 57 COMMUNITY PARK
PROPOSED ELECTRIC SERVICE

PREPARED FOR
OHIOVILLE BOROUGH COUNCIL
SITUATE IN
OHIOVILLE BOROUGH
BEAVER COUNTY, PENNSYLVANIA

